

PROJECT REPORT
ON
ANALYSIS OF BSNL'S COMPETITIVE POSITION IN THE INDIAN TELECOM
INDUSTRY

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DECLARATION

I hereby declare that the project report titled “**Analysis of BSNL’s Competitive Position in the Indian Telecom Industry** ” submitted for the **Master of Business Administration** is my original academic work prepared for the minor project requirement. The report is based on secondary research and has not formed the basis for the award of any degree, diploma, associateship, or other similar title. All ideas, data, and published findings used from external sources have been acknowledged in the text and listed in the references.

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ABSTRACT

India's telecom industry has changed a lot in the last few years because of the rapid growth of mobile internet, 4G and 5G services. Today, the market is mainly controlled by a few large private telecom companies. BSNL was once one of the leading telecom service providers in India, but over time it has lost a significant part of its wireless market share. Strong competition, slow technology upgrades and changing customer needs are some of the major reasons behind this decline.

This project focuses on understanding the current competitive position of BSNL in the Indian telecom industry with the help of business analytics. The study is based completely on secondary data collected from reliable sources such as TRAI reports, government publications, BSNL releases, Parliamentary documents and other published studies.

Simple analytical methods such as trend analysis, market share analysis, subscriber analysis, competitor comparison, circle-wise comparison and SWOT analysis have been used.

The study shows that BSNL has faced a decline in its subscriber base and market share over the years. Private telecom companies moved faster in adopting 4G and 5G technologies, while BSNL was comparatively late in upgrading its network. However, recent data also shows some improvement in its subscriber base and financial performance.

Overall, BSNL still has some important strengths such as government support, strong presence in rural areas, wide network reach and fiber infrastructure. For better future growth, BSNL needs to focus on faster technology upgrades, improved network quality, better customer service and effective use of business analytics for understanding and retaining customers.

INTRODUCTION

- In India, communication has become an essential aspect of daily life. It provides support for various activities such as digital payments, online education, healthcare, government services, business operations, and personal communication.
- Indian telecom has undergone significant changes over the years.? The shift from fixed-line services to 2G mobile networks was followed by 3G, 4G, and 5G. As of March 2026, India had over 1.28 billion wireless access subscriptions, as reported by TRAI, highlighting the telecom market's immense size.
- Currently, Reliance Jio is one of the major telecom operators, along with Bharti Airtel, Vodafone Idea, and BSNL (TRAI, 2026).
- The Department of Telecommunications service-related activities were transformed into a separate company, leading to the establishment of Bharat Sanchar Nigam Limited (BSNL) in 2000. BSNL's services have always been significant in the Indian telecom industry, as they extend beyond mobile connections. It provides various communication services such as landline, broadband, enterprise and government services, particularly in rural and remote regions. In 2002, BSNL launched 2G mobile services and in 2009, 3G services, suggesting that the company was once actively adopting new telecom technologies.
- However, the competitive situation changed with the rise of mobile internet and high-speed data. September 2016: Reliance Jio's entry into the market changed everything. By offering a 4G-based network and competitive pricing, it created an environment of increased competition and encouraged consumers to use their mobile data and digital services more frequently. With the launch of 5G services in India in October 2022, telecom operators placed greater emphasis on technology and network investment to enhance service delivery.

- Private telecom operators outpaced BSNL's 4G rollout throughout the country during this period, leading to problems. BSNL's 4G and 5G network advancement was reinforced by the Government of India's revival packages, which were also intended to enhance its standing. By 2026, the PIB showed that out of the 997,906 installed indigenous 4G sites, 96,103 were operational. BSNL's competitive position is the focus of this project, which utilizes business analytics.
- The article concentrates on examining subscriber patterns, market penetration and technology usage as well as financial performance, regional variations, and competition analysis. The main objective is to investigate the causes of BSNL's downfall, identify potential advancements and propose strategies that could assist in strengthening its presence in the Indian telecom sector.

OBJECTIVE OF THE STUDY

2.1 Statement of the Problem

- BSNL has faced a major decline in its competitive position in the wireless telecom market, especially during the period when mobile data became the main source of competition. According to TRAI data, BSNL's wireless subscriber base decreased from around 113.74 million in March 2022 to about 88.25 million in March 2024. Although the number later improved to nearly 92.94 million by March 2026, its wireless market share remained only 7.25%. In comparison, Reliance Jio had 39.71% and Bharti Airtel had 37.55% market share at the same time (TRAI, 2022; TRAI, 2026).
- One of the main reasons behind this decline is the slower adoption of new technology. Private telecom operators expanded their 4G networks much earlier and later moved quickly into 5G services. BSNL continued to depend mainly on older 2G and 3G networks for a longer period, while its nationwide 4G rollout was delayed. This affected its ability to provide high-speed data services when customer demand for internet-based services was increasing rapidly.
- BSNL has also faced financial pressure. Its total income increased from ₹18,595 crore in FY2020-21 to ₹23,427 crore in FY2024-25, while the annual net loss reduced from ₹7,441 crore to ₹2,247 crore. This shows improvement in financial performance, but the company still needs stronger market growth and better operational efficiency.
- Another issue is that BSNL's performance is not the same across all telecom circles. It has a stronger presence in some states and a weaker position in others. Therefore, the main problem studied in this project is to understand BSNL's present competitive position and identify the market, technology, financial and regional factors that can help improve its future performance.

2.2 Objectives of the Study

1. To study the trend of BSNL's wireless subscriber base.
2. To compare BSNL's market share with Jio, Airtel and Vodafone Idea.
3. To examine BSNL's recent financial and operational performance.
4. To compare BSNL's performance across selected telecom circles.
5. To understand the impact of delayed 4G and 5G adoption.
6. To identify BSNL's major strengths, weaknesses, opportunities and threats.
7. To suggest practical, data-based strategies for improving competitiveness and customer retention.

2.3 Scope of the Study

- The study focuses on the Indian telecom industry, mainly wireless access services. BSNL is the main company analysed, while Jio, Airtel and Vodafone Idea are used for comparison. Subscriber and market-share trends cover the period from March 2022 to March 2026. Financial analysis mainly covers FY2020-21 to FY2024-25. The project is completely based on secondary data and does not include any survey, interview or primary research.

2.4 Contribution of this Project

- This project brings together subscriber trends, market share, financial performance, technology adoption and telecom-circle analysis in one study. It helps explain not only how BSNL's position has changed, but also why the change happened. The project also shows that an increase in subscribers does not always mean an increase in market share if competitors are growing faster. These findings are used to develop practical suggestions for improving BSNL's competitive position.

REVIEW OF LITERATURE

3.1 Competition, Market Structure and the “Jio Effect”

- S. Parsheera and V. Trehan (2022) studied how the Indian telecom market changed after the entry of Reliance Jio. Their study showed that the number of active telecom operators reduced, data prices became cheaper and market concentration increased. This is important for the present study because BSNL’s decline happened during the same period. It suggests that BSNL’s weaker position was not only because of internal problems, but also because the overall telecom market became more competitive and concentrated.
- Mathews and Sivaraman (2025) analysed the Indian telecom industry using data from 2007 to 2023. They found that technology adoption and government regulation had an important effect on the market share of telecom companies. Their findings are useful for this project because BSNL’s delayed adoption of 4G and 5G is one of the major factors being studied.
- Kumar and Bose (2022) discussed the challenges faced by BSNL and MTNL. They highlighted issues such as management limitations, financial pressure, changing customer expectations and the need for better business strategies. However, their study was conducted before BSNL’s recent 4G rollout and financial improvement, so an updated analysis is needed.

3.2 Customer Switching and Technology Adoption

- Dhankhar and Dasgupta (2023) studied the reasons behind customers switching telecom brands in India. They found that customers compare operators based on price, service quality and value-added services. This is relevant to BSNL because losing subscribers can also be understood as a customer retention problem. Customers today expect better internet speed, network quality, digital services and overall convenience.

- TRAI's Yearly Performance Indicators for 2024-25 showed that Jio and Airtel were the two largest wireless operators, while BSNL's market share remained below 8%. TRAI's January–March 2026 report further showed that BSNL had around 92.94 million wireless subscribers and a 7.25% market share. These reports provide important official data for comparing BSNL with its competitors.

3.3 Revival Policy and Financial Performance

- The Ministry of Communications (PIB, 2023) reported a revival package of ₹89,047 crore for BSNL, mainly for 4G and 5G spectrum. Earlier revival packages were also introduced to support debt restructuring, rural connectivity and network expansion.
- According to PIB, BSNL reported total income of ₹23,427 crore in FY2024-25, while its annual net loss reduced to ₹2,247 crore. The company also reported profits in the third and fourth quarters of the year. A Rajya Sabha replies in March 2026 also showed that BSNL's income improved and losses reduced over the five-year period from FY2020-21 to FY2024-25.

3.4 Network Modernisation and Research Gap

- According to the BSNL Revival Roadmap (PIB, 2026), 97,906 indigenous 4G sites had been installed by 28 February 2026, and 96,103 were operational. This shows that BSNL is now improving its network, although the rollout came much later than private competitors.
- Overall, previous studies explain market competition, customer switching, technology adoption and BSNL's financial challenges. However, there is limited research that combines subscriber trends, market share, financial performance, technology development and circle-wise analysis in one updated study. This project attempts to fill that gap by using recent secondary data and business analytics methods to understand BSNL's present competitive position and possible areas for improvement.

RESEARCH METHODOLOGY

4.1 Research Design

- This study follows a descriptive, exploratory and comparative research design. The descriptive approach is used to present BSNL's subscriber base, market share, financial performance and telecom-circle position. The exploratory approach helps in understanding the possible reasons behind the change in BSNL's competitive position over time. The comparative approach is used to compare BSNL with major telecom competitors such as Reliance Jio, Bharti Airtel and Vodafone Idea. Historical trend analysis is also used to understand how BSNL's performance has changed between March 2022 and March 2026.

4.2 Sources of Data Collection

- The project is completely based on secondary data. No survey, questionnaire or interview has been conducted. The main sources of data include reports published by the Telecom Regulatory Authority of India (TRAI), Ministry of Communications, Press Information Bureau (PIB), Parliamentary replies, official company releases and selected academic studies.
- These sources were selected because they provide reliable information about telecom subscribers, market share, service areas, financial performance, technology development and government support for BSNL. Using official sources also helps in keeping the analysis more accurate and relevant.

4.3 Sampling Design and Technique

- Since the study is based on secondary data, there is no sample of individual respondents. A purposive or judgmental sampling method is used to select the most relevant datasets for the analysis. BSNL, Reliance Jio, Bharti Airtel and Vodafone Idea are selected for national-level comparison because they are the main wireless telecom operators in India.

- For trend analysis, March year-end data from 2022 to 2026 are mainly used to maintain consistency. Fourteen selected telecom service areas are also used for regional comparison based on the availability of comparable March 2026 data. These telecom circles do not always follow exact state boundaries. For example, the Bihar telecom circle includes Jharkhand, while the Andhra Pradesh circle also includes Telangana under TRAI classification.

4.4 Tools and Framework for Data Analysis

- The collected data are analysed mainly using simple business analytics methods. Microsoft Excel-style calculations are used for percentages, changes and comparisons. Trend analysis is used to study changes in BSNL's subscriber base and market share. CAGR is used where a multi-year growth rate is useful.
- Competitor benchmarking is used to compare subscriber size, market share, technology position and rural presence. Circle-wise analysis is used to identify areas where BSNL has a relatively stronger or weaker market position. Financial trend analysis is used to study total income and net profit or loss. SWOT analysis is also used to identify BSNL's major strengths, weaknesses, opportunities and threats.
- Although TRAI monthly data were available up to June 2026, March 2026 data are mainly used for detailed operator and circle comparison because the quarter-end report provides a consistent dataset for all service areas.

DATA ANALYSIS AND INTERPRETATION

The data collected from TRAI, government reports and other official sources were analysed to understand BSNL's present position in the Indian telecom market. The analysis mainly focuses on market share, subscriber trends, financial performance, technology development, regional performance and comparison with major competitors.

5.1 Indian Telecom Market Share

- The Indian wireless telecom market is mainly dominated by Reliance Jio and Bharti Airtel. At the end of March 2026, Jio had the highest wireless access market share at **39.71%**, followed by Airtel at **37.55%**. Vodafone Idea had **15.48%**, while BSNL held only **7.25%**. Together, Jio and Airtel controlled more than 77% of the market. This shows that BSNL is competing in a highly concentrated market where the two largest private operators have a major advantage in subscriber size, technology and network investment (TRAI, 2026).

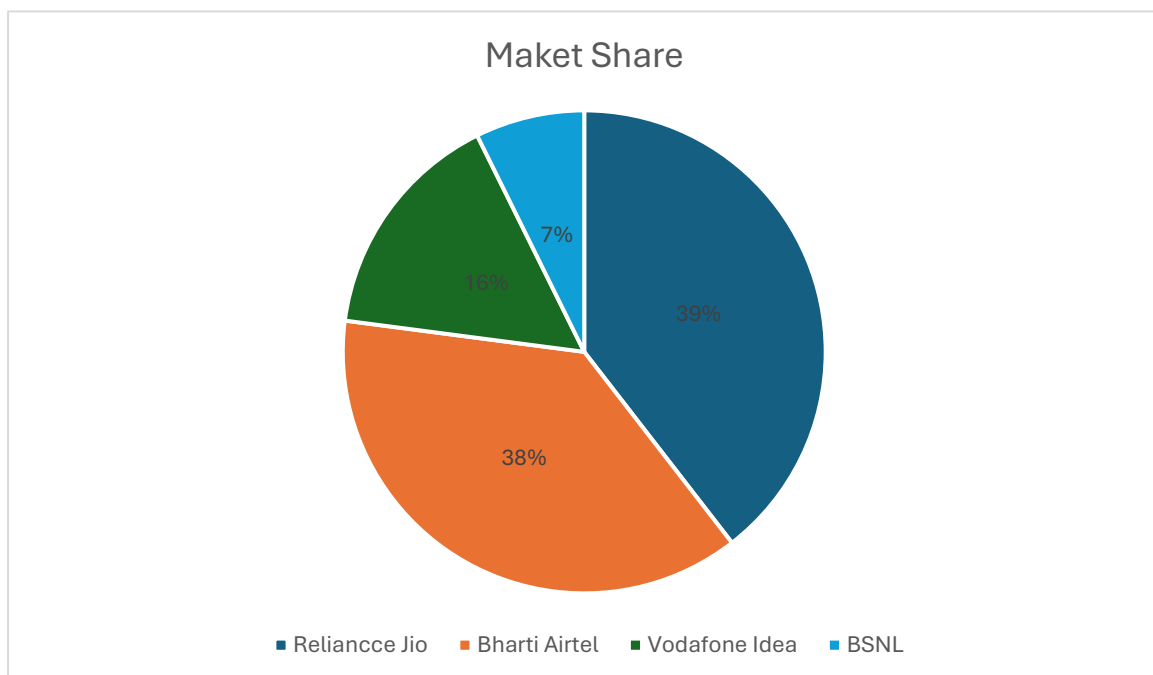


Figure 1: Wireless Access Market Share of Major Operators, March 2026

5.2 BSNL Subscriber Trend

- BSNL's subscriber trend shows both decline and some recent recovery. Its wireless subscriber base was around **113.74 million in March 2022**, which declined to **88.25 million in March 2024**. After this, the subscriber base improved to **91.06 million in March 2025** and further increased to approximately **92.94 million in March 2026**. This recovery is a positive sign, but BSNL still had about 20.8 million fewer subscribers than in March 2022. Therefore, the recent growth can be considered a sign of stabilisation rather than a complete recovery.

5.3 Comparison with Major Competitors

- In March 2026, BSNL had around **92.94 million wireless subscribers**, compared with **509.21 million for Jio, 481.48 million for Airtel and 198.48 million for Vodafone Idea**. BSNL also had around **30.80 million rural wireless subscribers**. Its main strengths are its government support, rural infrastructure, fibre network and strategic role. However, Jio and Airtel have a clear advantage in subscriber scale, 4G/5G services and digital platforms.

5.4 Financial Performance

- BSNL's financial data show improvement in recent years. Total income increased from **₹18,595 crore in FY2020-21 to ₹23,427 crore in FY2024-25**. During the same period, its annual net loss reduced from **₹7,441 crore to ₹2,247 crore**. Although BSNL was still making an annual loss, the reduction in losses and increase in income indicate better financial performance and operational improvement.

5.5 Telecom-Circle Analysis

- BSNL's market position is different across telecom circles. In March 2026, its share was strongest among the selected circles in **Kerala at 20.97%**, followed by **Odisha at 14.24% and Punjab at 11.23%**. In comparison, its share was only around **4.51% in West Bengal and 4.55% in Gujarat**. This difference shows that BSNL should not follow the same strategy in every region. Strong circles can focus more on customer retention and 4G/FTTH upgrades, while weaker circles need improvement in network quality and customer acquisition.

5.6 Overall Interpretation

- Overall, the analysis shows that BSNL still faces a large competitive gap with Jio and Airtel, mainly because of slower 4G adoption, lower subscriber scale and weaker digital services. However, its recent subscriber recovery, improving financial performance, expanding 4G network and strong rural and fibre infrastructure provide opportunities for future growth. A SWOT analysis also suggests that government backing and infrastructure are major strengths, while delayed technology adoption, customer churn and strong private competition remain important challenges.

RESULTS AND DISCUSSION

6.1 Findings of the Study

- The analysis shows that BSNL is currently a much smaller player in the Indian wireless telecom market compared with Jio and Airtel. In March 2026, BSNL had a market share of **7.25%**, while Jio and Airtel had **39.71%** and **37.55%** respectively. This clearly shows the wide competitive gap between BSNL and the two leading private operators.
- The subscriber trend also shows that BSNL faced a major decline between 2022 and 2024. Its wireless subscriber base reduced from around **113.74 million in March 2022 to 88.25 million in March 2024**. However, the number improved to about **92.94 million by March 2026**. This recovery is a positive sign, but BSNL has still not returned to its earlier subscriber level. Its market share also declined from around **9.96% in March 2022 to 7.25% in March 2026**, which means that competitors grew faster during the same period.
- Technology delay appears to be one of the major reasons behind this decline. BSNL introduced 2G and 3G services at an early stage, but its nationwide 4G rollout happened much later than private operators. By the time BSNL started expanding 4G on a large scale, Jio and Airtel had already developed strong 4G customer bases and moved towards 5G services. Still, BSNL's recent 4G development is an important sign of improvement. By February 2026, more than **96,000 4G sites were operational**.
- Financial performance has also improved in recent years. BSNL's total income increased to **₹23,427 crore in FY2024-25**, while its annual net loss reduced to **₹2,247 crore**. The company also reported profits in the third and fourth quarters of the year. This indicates improvement, although full-year profitability has not yet been achieved.

- The circle-wise analysis shows that BSNL's performance is different across India. Kerala, Odisha and Punjab have a relatively stronger BSNL presence, while its share is much lower in Gujarat, West Bengal, Maharashtra and Bihar. Therefore, one common strategy may not work equally well in every telecom circle.

6.2 Suggestions

- Based on the findings, BSNL should first focus on converting its growing 4G network into a better customer experience. Network speed, coverage, complaints, site utilisation and customer additions should be regularly monitored at the circle level.
- BSNL should also prepare a clear and time-based 5G expansion plan, especially for areas where demand is high. Customer retention can be improved by using business analytics to identify customers who are likely to leave based on recharge behaviour, complaints and usage patterns.
- Different circles should have different strategies. Stronger markets such as Kerala and Odisha can focus on retaining and upgrading existing customers, while weaker circles need better network quality and targeted marketing.
- BSNL should also improve its mobile app, digital support and complaint-resolution process. Bundling mobile services with Bharat Fibre, FTTH, enterprise and government connectivity can create additional value. Overall, BSNL's revival is visible, but long-term competitiveness will depend on faster execution, better service quality and stronger use of data in business decisions.

CONCLUSION

- BSNL has played an important role in the development of telecommunications in India, especially in rural connectivity, landline services, government networks and public communication services. However, the Indian telecom industry has changed rapidly with the growth of mobile data, 4G and 5G services. Strong competition from private operators, especially Jio and Airtel, has created major challenges for BSNL.
- The study shows that BSNL's wireless subscriber base and market share declined significantly over the years. Although the subscriber base recovered to around **92.94 million by March 2026**, its market share remained only **7.25%**. This means that BSNL is showing some recovery, but competitors are still growing at a much faster rate.
- One of the main reasons behind BSNL's weaker position was the delay in nationwide 4G expansion. Private operators had already developed large 4G networks and moved towards 5G before BSNL completed its major 4G rollout. However, the activation of more than **96,000 4G sites by February 2026** is an important step towards improving its network.
- BSNL has also shown improvement in its financial performance. Its total income has increased; losses have reduced and the company reported profits in two consecutive quarters during FY2024-25. Government support, fibre infrastructure, rural presence and future 5G capability also provide opportunities for growth.
- Overall, BSNL's revival has started, but it is not complete. Its future success will depend on faster technology implementation, better network quality, improved customer service and effective use of business analytics. If BSNL can use data to understand customer needs and improve regional performance, it can strengthen its competitive position in the Indian telecom market.

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